



Robotics and Automation Engineering

2025//2026

PROJECT INDUSTRIAL AUTOMATION

PRESENTED BY:

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Index

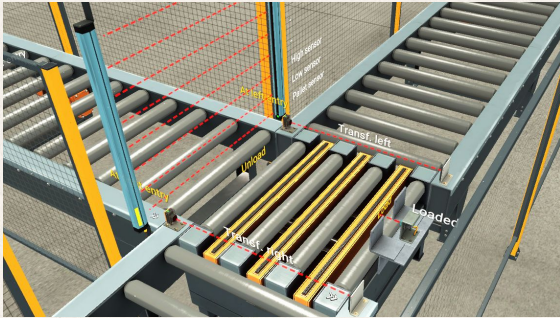
01	Overview
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Overview

Introduce to the project.

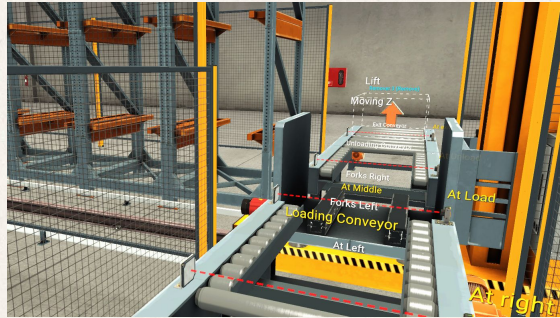
Transfer

Transfer the package either to the left or to the right.



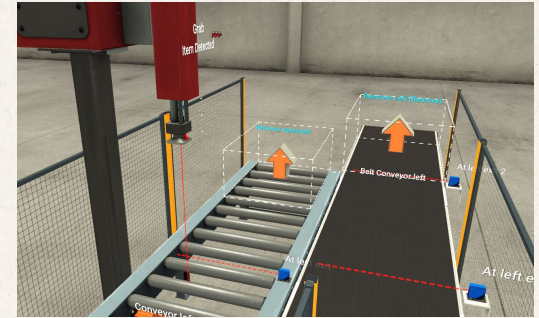
Warehouse

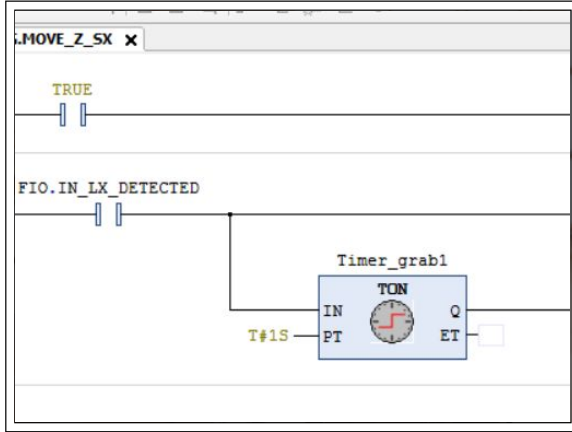
Small package are sent to a warehouse.



Pick & Place

Higher package are picked and place into a belt conveyor.





01 Robustness and Reliability

PLCs are designed to work in harsh industrial environments and ensure high reliability.

02 Determinism

A PLC executes the program with predictable cycle times, allowing precise and real-time control of processes.

03 Modularity and Scalability

PLCs allow easy addition or modification of input/output modules, making the system expandable according to needs.

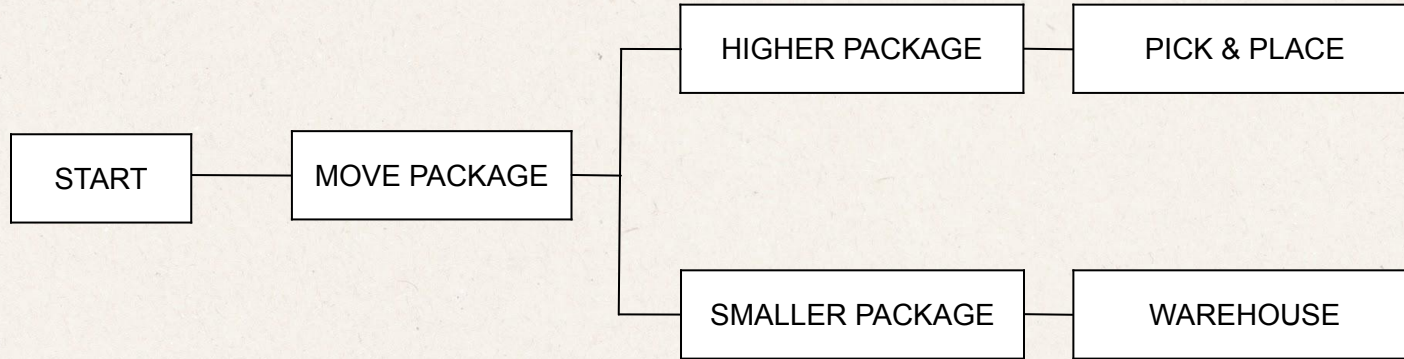
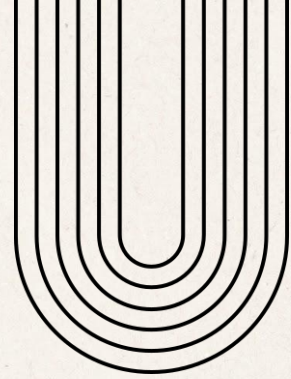
Why PLC?

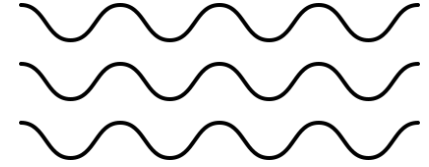
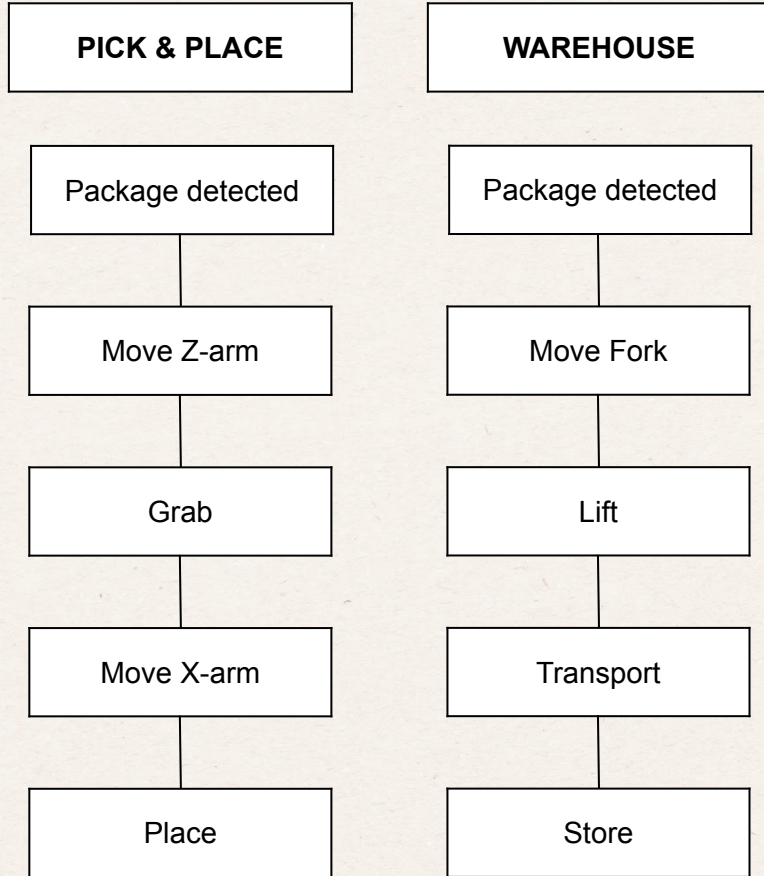
PLCs are widely used in industrial automation to efficiently control and coordinate machines and processes.



Process Flowchart

The logic behind the project.





Flowchart

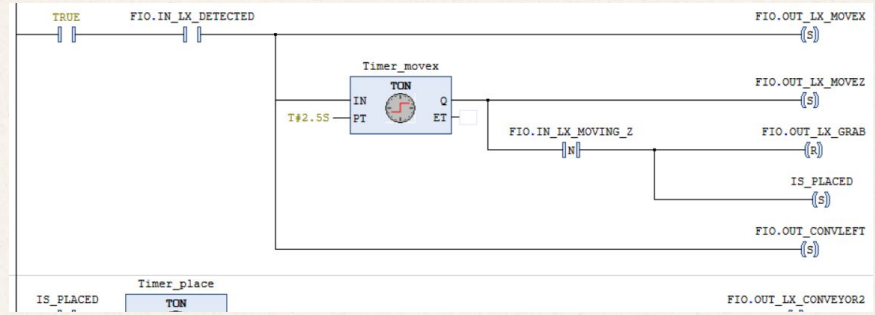
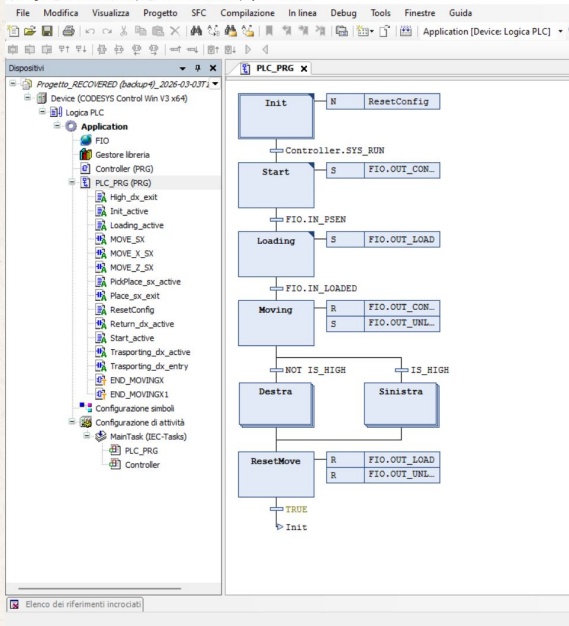
How do the main tasks work?

Sensors and Actuators

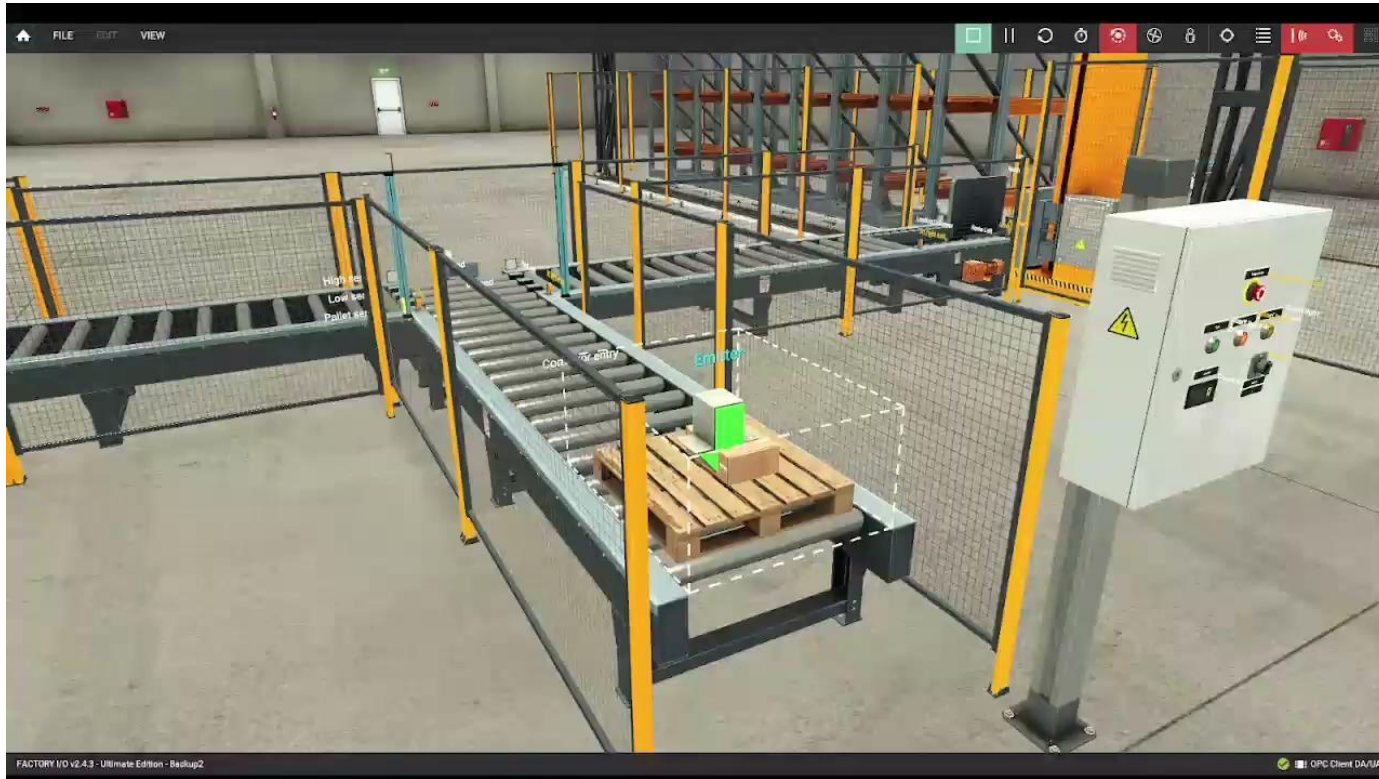
At Left	IN_DX_LEFT
At Load	IN_DX_LOAD
At Middle	IN_DX_MIDDLE
Moving X	IN_DX_MOVING_X
Moving Z	IN_DX_MOVING_Z
At Right	IN_DX_RIGHT
At Unload	IN_DX_UNLOAD
Emergency stop	...EMERGENCYSTOP
High sensor	IN_HIGHSEN
At left entry	IN_LENTRY
At left exit	IN_LEXIT
Loaded	IN_LOADED
Low sensor	IN_LOWSEN
Item Detected	IN_LX_DETECTED
At left entry 2	IN_LX_LENTRY2
At left exit 2	IN_LX_LEXIT2
Moving X (sx)	IN_LX_MOVING_X
Moving Z (sx)	IN_LX_MOVING_Z
Pallet sensor	IN_PSEN
At right entry	IN_RENTRY
Reset	IN_RESET
At right exit	IN_REXIT
Start	IN_START
Stop	IN_STOP

Stop	IN_STOP	
	OUT_CONVENTRY	Conveyor entry
	OUT_CONVLEFT	Conveyor left 1
	OUT_CONVRIGHT	Conveyor right
	OUT_DX_LFORK	Forks Left
	OUT_DX_LIFT	Lift
	OUT_DX_LOADCONV	Loading Conveyor
	OUT_DX_RFORK	Forks Right
	OUT_DX_TARGETPOS	Target Position
	...DX_UNLOADCONV	Unloading Conveyor
	OUT_EMITTER	
	OUT_LOAD	Load
	OUT_LTRANS	Transf. left
	OUT_LX_CONVEYOR2	Belt Conveyor left
	OUT_LX_GRAB	Grab
	OUT_LX_MOVEX	Move X (sx)
	OUT_LX_MOVEZ	Move Z (sx)
	OUT_RESETLIGHT	Reset light
	OUT_RTRANS	Transf. right
	OUT_STARTLIGHT	Start light
	OUT_STOPLIGHT	Stop light
	OUT_UNLOAD	Unload
	OnlineAccess	

Codesys



Video - Factory IO





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Thank you
